



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 2)

2015 HSC Course Assessment Task 2

Thursday March 5, 2015

General instructions

- Working time – 55 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 6)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Lam

☐ 12M4B – Mr Lin

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-5	6	7	8	Total	%
MARKS	$\overline{5}$	$\overline{11}$	$\overline{13}$	$\overline{12}$	$\overline{41}$	

Section I

5 marks

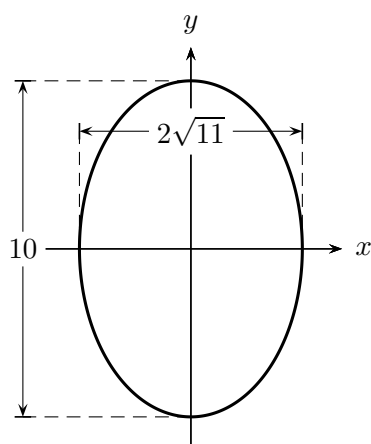
Attempt Question 1 to 5

Mark your answers on the answer grid provided.

Questions

Marks

1. A conic section is graphed as shown. The distance between the x intercepts is $2\sqrt{11}$ units, and the distance between the y intercepts is 10 units. Which of the following is the equation of the graph? 1



(A) $\frac{x^2}{25} + \frac{y^2}{44} = 1$

(B) $\frac{x^2}{44} + \frac{y^2}{25} = 1$

(C) $\frac{x^2}{25} + \frac{y^2}{11} = 1$

(D) $\frac{x^2}{11} + \frac{y^2}{25} = 1$

2. Which of the following represents the equation of the chord of contact from $(5, -2)$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$? 1

(A) $\frac{x}{8} - \frac{5y}{16} = 1$ (B) $\frac{5x}{16} + \frac{2y}{9} = 1$ (C) $\frac{5x}{16} - \frac{2y}{9} = 0$ (D) $\frac{5x}{16} - \frac{2y}{9} = 1$

3. $S(4, 0)$ is a focus of the rectangular hyperbola $x^2 - y^2 = k$. Which of the following is the value of k ? 1

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 8 (D) 32

4. Which of the following is the equation of the conic represented by foci $(\pm 3, 0)$ and vertices $(\pm 2, 0)$? 1

(A) $\frac{x^2}{5} + \frac{y^2}{4} = 1$ (B) $\frac{x^2}{4} + \frac{y^2}{5} = 1$ (C) $\frac{x^2}{4} - \frac{y^2}{5} = 1$ (D) $\frac{x^2}{5} - \frac{y^2}{4} = 1$

5. If $3x^2 + 2xy + y^2 = 2$, what is the gradient of the tangent at $x = 1$? 1

(A) -2 (B) 0 (C) 4 (D) not defined

Section II

36 marks

Attempt Questions 6 to 8

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (11 Marks)	Commence a NEW page.	Marks
------------------------------	----------------------	--------------

- | | | |
|-----|--|----------|
| (a) | Consider the hyperbola $\mathcal{H}$ with equation $\frac{x^2}{9} - \frac{y^2}{16} = 1$. | |
| | i. Find the points of intersection of $\mathcal{H}$ with the x axis, as well as the eccentricity and foci of $\mathcal{H}$. | 3 |
| | ii. Write down the equations of the directrices and asymptote of $\mathcal{H}$. | 2 |
| | iii. Sketch $\mathcal{H}$, showing all of the above information. | 2 |
| (b) | i. Expand and simplify: $(1 + \sqrt{2})^2$. | 1 |
| | ii. A sequence a_n is defined by $a_0 = a_1 = 2$, and | 3 |

$$a_n = 2a_{n-1} + a_{n-2}$$

when $n \geq 2$.

Use mathematical induction to prove that

$$a_n = (1 + \sqrt{2})^n + (1 - \sqrt{2})^n \quad \text{for all } n \geq 0$$

Question 7 (13 Marks)	Commence a NEW page.	Marks
------------------------------	----------------------	--------------

- | | | |
|-----|---|----------|
| (a) | Describe in geometric terms, the locus of a point z which moves on the Argand diagram according to the equation | 3 |
|-----|---|----------|

$$|z - 3| + |z + 3| = 8$$

State the coordinates of the focus and equation of the directrices.

- | | | |
|-----|--|----------|
| (b) | The hyperbola $\mathcal{B}$ has equation $xy = 4$. | |
| | i. Sketch $\mathcal{B}$, indicating on your diagram the positions and coordinates of all points at which $\mathcal{B}$ intersects the axes of symmetry. | 2 |
| | ii. Show that the equation of the tangent to $\mathcal{B}$ at $P\left(2t, \frac{2}{t}\right)$ where $t \neq 0$, is | 2 |

$$x + t^2y = 4t$$

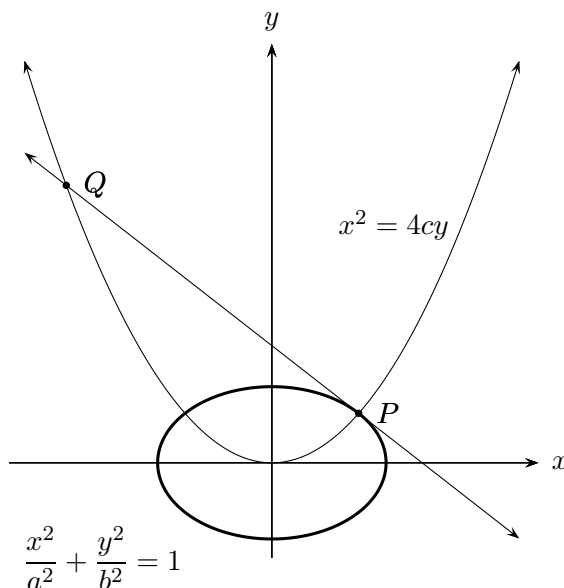
- | | | |
|------|---|----------|
| iii. | If $s \neq 0$ and $s^2 \neq t^2$, show that the tangents to $\mathcal{B}$ at P and $Q\left(2s, \frac{2}{s}\right)$ intersect at $M\left(\frac{4st}{s+t}, \frac{4}{s+t}\right)$. | 3 |
| iv. | Suppose that in the previous part, the parameter $s = -\frac{1}{t}$. Show that the locus of M is a straight line through, but excluding the origin. | 3 |

Question 8 (12 Marks)

Commence a NEW page.

Marks

The parabola $x^2 = 4cy$ intersects the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(2cp, cp^2)$ in the first quadrant where $a > b > 0$ and $c > 0$. The tangent to the ellipse at P meets the parabola again at $Q(2cq, cq^2)$.



- (a) Show that the tangent to the ellipse at P is given by

3

$$\frac{2cpx}{a^2} + \frac{cp^2y}{b^2} = 1$$

- (b) If this tangent meets the parabola at $(2ct, ct^2)$, show that $\frac{p^2t^2}{b^2} + \frac{4pt}{a^2} - \frac{1}{c^2} = 0$ and deduce that, considered as a quadratic in t , this equation has roots p and q .

2

- (c) If PQ subtends a right angle at the origin, show that $pq = -4$ and deduce that

3

$$\frac{1}{b^2} = \frac{1}{a^2} + \frac{1}{(4c)^2}$$

- (d) Using the information from parts (b) and (c) or otherwise, show that if PQ subtends a right angle at the origin, then $p = 2e$, where e is the eccentricity of the ellipse.

4**End of paper.**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

Section I

(Lam) 1. (D) 2. (D) 3. (C) 4. (C) 5. (D)

Section II

Question 6 (Ireland)

(a) i. (3 marks)

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

Semi major axis is 3, hence

$$V(\pm 3, 0)$$

Eccentricity:

$$\begin{aligned} b^2 &= a^2 (e^2 - 1) \\ 16 &= 9 (e^2 - 1) \\ e^2 - 1 &= \frac{16}{9} \\ \therefore e^2 &= \frac{16}{9} + 1 = \frac{25}{9} \\ e &= \frac{5}{3} \end{aligned}$$

Foci:

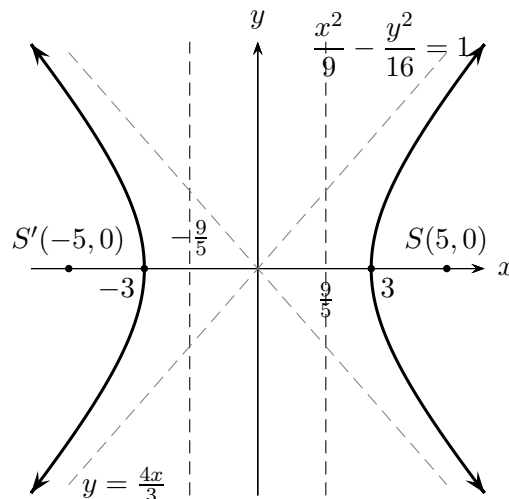
$$S(\pm ae, 0) = \left(\pm 3 \times \frac{5}{3}, 0 \right) = (\pm 5, 0)$$

ii. (2 marks)

$$\text{Directrices: } x = \pm \frac{a}{e} = \pm \frac{3}{\frac{5}{3}} = \pm \frac{9}{5}$$

$$\text{Asymptotes: } y = \pm \frac{b}{a}x = \pm \frac{4x}{3}$$

iii. (2 marks)



(b) i. (1 mark)

$$(1 + \sqrt{2})^2 = 1 + 2\sqrt{2} + 2 = 3 + 2\sqrt{2}$$

ii. (3 marks)

Let $P(n)$ be the proposition

$$a_n = (1 + \sqrt{2})^n + (1 - \sqrt{2})^n$$

whenever $n \geq 0$, with $a_0 = a_1 = 2$ and

$$a_n = 2a_{n-1} + a_{n-2}$$

• Base cases: $P(0)$ and $P(1)$

– $P(0)$:

$$a_0 = (1 + \sqrt{2})^0 + (1 - \sqrt{2})^0 = 2$$

– $P(1)$:

$$\begin{aligned} a_1 &= (1 + \sqrt{2})^1 + (1 - \sqrt{2})^1 \\ &= 1 + \sqrt{2} + 1 - \sqrt{2} = 2 \end{aligned}$$

Hence $P(1)$ and $P(2)$ are true.

• Inductive hypotheses: assume $P(3), P(4), \dots, P(k)$ and $P(k+1)$ are all true, i.e.

– $P(k)$:

$$a_k = (1 + \sqrt{2})^k + (1 - \sqrt{2})^k$$

– $P(k+1)$:

$$a_{k+1} = (1 + \sqrt{2})^{k+1} + (1 - \sqrt{2})^{k+1}$$

Examine $P(k+2)$:

$$\begin{aligned} a_{k+2} &= 2a_{k+1} + a_k \\ &= 2 \left[(1 + \sqrt{2})^{k+1} + (1 - \sqrt{2})^{k+1} \right] + (1 + \sqrt{2})^k + (1 - \sqrt{2})^k \\ &= (1 + \sqrt{2})^k (2(1 + \sqrt{2}) + 1) + (1 - \sqrt{2})^k (2(1 - \sqrt{2}) + 1) \\ &= (1 + \sqrt{2})^k (3 + 2\sqrt{2}) + (1 - \sqrt{2})^k (3 - 2\sqrt{2}) \\ &= (1 + \sqrt{2})^k (1 + \sqrt{2})^2 + (1 - \sqrt{2})^k (1 - \sqrt{2})^2 \\ &= (1 + \sqrt{2})^{k+2} + (1 - \sqrt{2})^{k+2} \end{aligned}$$

Hence $P(n)$ is true by induction.

Question 7 (Lin)

(a) (3 marks)

- ✓ [1] for each of description of locus, foci and directrices.

$$|z - 3| + |z + 3| = 8$$

$$\Downarrow$$

$$PS + PS' = 2a$$

$$\therefore S(\pm 3, 0) \quad a = 4$$

Equation of directrices: found from eccentricity.

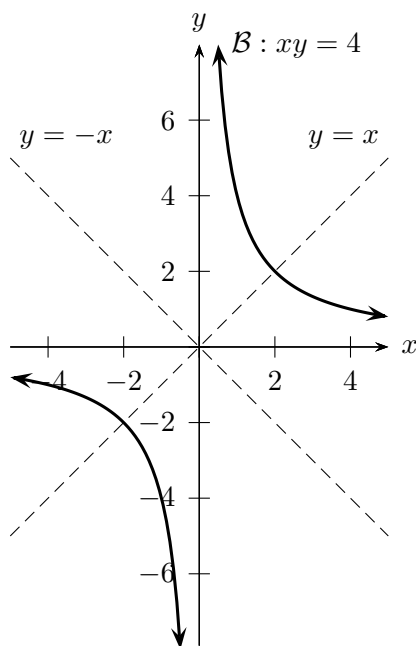
$$ae = 3$$

$$4e = 3$$

$$\therefore e = \frac{3}{4}$$

$$\therefore x = \pm \frac{a}{e} = \frac{4}{\frac{3}{4}} = \frac{16}{3}$$

(b) i. (2 marks)



Intersection with axes of symmetry:

$$\begin{cases} y = \frac{4}{x} \\ y = x \end{cases}$$

$$\therefore -x = \frac{4}{x}$$

$$x^2 = 4$$

$$x = \pm 2 \quad y = \pm 2$$

ii. (2 marks)

Equation of tangent at $(2t, \frac{2}{t})$:

$$y = 4x^{-1}$$

$$\frac{dy}{dx} = -4x^{-2} \Big|_{x=2t} = \frac{-4}{4t^2} = -\frac{1}{t^2}$$

Apply the point-gradient formula,

$$\frac{y - \frac{2}{t}}{x - 2t} = -\frac{1}{t^2}$$

$$t^2 y - 2t = -x + 2t$$

$$x + t^2 y = 4t$$

iii. (3 marks)

At S , the equation of the tangent is $x + s^2 y = 4s$. Where the tangents from P And Q intersect,

$$\begin{cases} x + t^2 y = 4t & (1) \\ x + s^2 y = 4s & (2) \end{cases}$$

(1) - (2):

$$y(t^2 - s^2) = 4t - 4s$$

$$y(t - s)(t + s) = 4(t - s)$$

$$\therefore y = \frac{4}{s + t}$$

Substitute into (1) to find x :

$$x + t^2 \left(\frac{4}{s + t} \right) = 4t$$

$$x + \frac{4t^2}{s + t} = \frac{4t(s + t)}{s + t}$$

$$x = \frac{4st + 4t^2 - 4t^2}{s + t} = \frac{4st}{s + t}$$

Hence the coordinates of M are

$$M \left(\frac{4st}{s + t}, \frac{4}{s + t} \right)$$

iv. (3 marks)

- ✓ [1] for each of x_M and y_M

- ✓ [1] for $y = -x$ and justification for $x \neq 0$.

Spse $s = -\frac{1}{t}$. Hence $t \neq 0$ and $s \neq 0$ and $x \neq 0$.

$$\begin{cases} x_M = \frac{4st}{s + t} = \frac{4 \times -\frac{1}{t} \times t}{-\frac{1}{t} + t} = -\frac{4}{t - \frac{1}{t}} \\ y_M = \frac{4}{s + t} = \frac{4}{t - \frac{1}{t}} \end{cases}$$

Hence $y_M = -x_M$ and the locus of M is on the line $y = -x$, excluding $(0, 0)$ as $t \neq 0$.

Question 8 (Lam)

(a) (3 marks)

- Equation of tangent to ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Differentiating to obtain gradient at (x_0, y_0) :

$$\begin{aligned} \frac{d}{dx} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) &= 0 \\ \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{b^2 x}{a^2 y} \bigg|_{\substack{x=x_0 \\ y=y_0}} \\ &= -\frac{b^2 x_0}{a^2 y_0} \end{aligned}$$

Point-gradient formula

$$\begin{aligned} \frac{y - y_0}{x - x_0} &= -\frac{b^2 x_0}{a^2 y_0} \\ a^2 y y_0 - a^2 y_0^2 &= -b^2 x x_0 + b^2 x_0^2 \\ \underbrace{b^2 x x_0 + a^2 y y_0}_{\div a^2 b^2} &= \underbrace{b^2 x_0^2 + a^2 y_0^2}_{\div a^2 b^2} \\ \frac{x x_0}{a^2} + \frac{y y_0}{b^2} &= \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} \end{aligned}$$

As (x_0, y_0) is a point on the ellipse, then it satisfies $\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$:

$$\therefore \frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1$$

- As $x_0 = 2cp$, $y_0 = cp^2$ (a point on the parabola also), the equation of the tangent is

$$\frac{2cpx}{a^2} + \frac{cp^2 y}{b^2} = 1$$

(b) (2 marks)

Solving equation of tangent to ellipse & equation of parabola simultaneously:

$$\begin{cases} \frac{2cpx}{a^2} + \frac{cp^2 y}{b^2} = 1 & (1) \\ x^2 = 4cy & (2) \end{cases}$$

Rearrange (2) and substitute into (1):

$$\begin{aligned} y &= \frac{x^2}{4c} \\ \frac{2cpx}{a^2} + \frac{cp^2 \left(\frac{x^2}{4c} \right)}{b^2} &= 1 \end{aligned}$$

At $x = 2ct$,

$$\begin{aligned} \frac{2cp(2ct)}{a^2} + \frac{cp^2 (2ct)^2}{4b^2} &= 1 \\ \frac{4c^2 pt}{a^2} + \frac{4p^2 t^2 c^2}{b^2} &= 1 \end{aligned}$$

Divide throughout by c^2 ,

$$\begin{aligned} \frac{4pt}{a^2} + \frac{p^2 t^2}{b^2} &= \frac{1}{c^2} \\ \therefore \frac{p^2 t^2}{b^2} + \frac{4pt}{a^2} - \frac{1}{c^2} &= 0 \quad (\spadesuit) \end{aligned}$$

Quadratic in t will have 2 roots if it were to intersect the parabola twice. Those points of intersection are at $P(2cp, cp^2)$ and $Q(2cq, cq^2)$. At those points, the parameter $t = p$ or $t = q$. Hence the roots to the quadratic are p and q .

(c) (3 marks)

- Gradient of OP : $m_{OP} = \frac{cp^2 - 0}{2cp - 0} = \frac{p}{2}$. Similarly,

$$m_{OQ} = \frac{q}{2}$$

- As PQ subtends a right angle at O , then $OP \perp OQ$:

$$\begin{aligned} m_{OP} \times m_{OQ} &= -1 \\ \frac{p}{2} \times \frac{q}{2} &= -1 \\ \therefore pq &= -4 \end{aligned}$$

- From the previous quadratic in t , if $t = q$ then

$$\begin{aligned} \frac{p^2 q^2}{b^2} + \frac{4pq}{a^2} - \frac{1}{c^2} &= 0 \\ \frac{16}{b^2} + \frac{-16}{a^2} - \frac{1}{c^2} &= 0 \\ \therefore \frac{1}{b^2} &= \frac{1}{a^2} + \frac{1}{16c^2} \\ \therefore \frac{1}{b^2} &= \frac{1}{a^2} + \frac{1}{(4c^2)} \quad (\dagger) \end{aligned}$$

(d) (4 marks)

- Given $pq = -4$, and p, q being roots to the quadratic in t ($\spadesuit$), then the product of roots is -4 :

$$pq = \frac{-\frac{1}{c^2}}{\frac{p^2}{b^2}} = -4$$

$$\therefore -\frac{b^2}{c^2 p^2} = -4$$

$$\frac{b^2}{4c^2} = p^2$$

$$\therefore p = \frac{b}{2c}$$

- Also, $b^2 = a^2(1 - e^2)$:

$$\therefore \frac{b^2}{a^2} = 1 - e^2$$

- Take equation ($\ddagger$) and multiply throughout by b^2 :

$$1 = \frac{b^2}{a^2} + \frac{b^2}{(4c)^2}$$

$$1 = (1 - e^2) \frac{b^2}{16c^2}$$

$$\therefore e^2 = \frac{b^2}{16c^2}$$

$$e = \frac{b}{4c} = \frac{1}{2} \times \frac{b}{2c} = \frac{1}{2}p$$

$$\therefore p = 2e$$